AUMM 9/1/2026 :-

- India → Thick seam = 4.8 m.
- Ground control measures needed to be done.

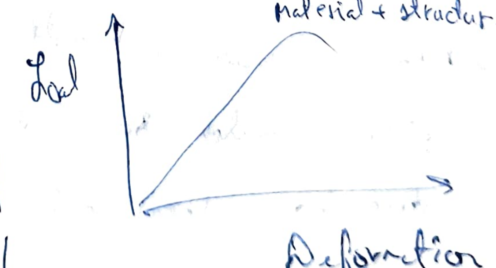
- B & P → pillar crush, vent issue.
- LW → better option.
- IPCC → contain within 1.5 °C., change in temperature
- O/C → SR, maintainance of slope angle, high waste handle cost,
  high slope → less waste, but stability issues.
- Beyond 300 m depth & thick seams.

★ Ground Control & Rock Mechanics Aspect of Thick Seam Mining.

- W/H ratio.
- Natural pressure arch.
- Pillar Spalling.
- en-masse caving.
- natural caving.
- Caving angle.
- abutment angle.

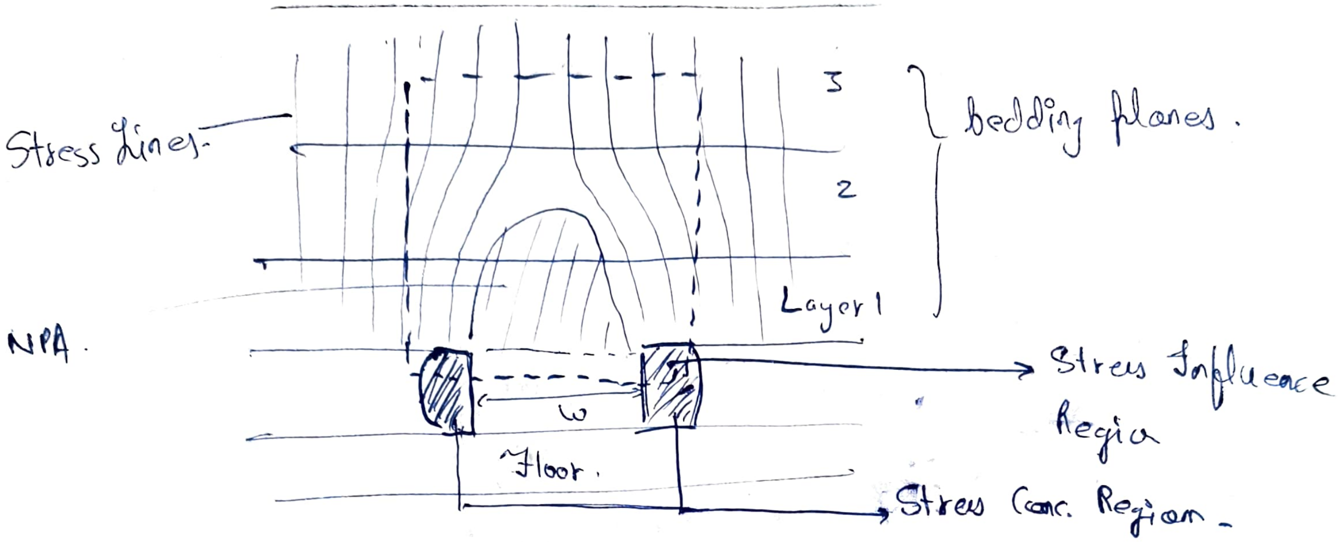
① Same w_1
→ $h_1 < h_2 < h_3$
⟸
↑ in pillar strength & R ↓
→ stiffness parameter.

material + structural prop



→ modulus → material prop

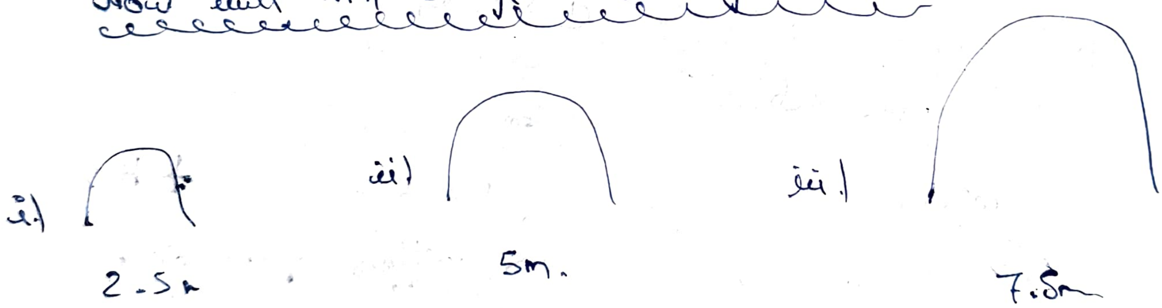
② Natural Pressure Arch.



- NPA → Stress Relaxed Zone, Pressure Arch.
  → Zero load capacity., it transfers to surrounding. { stress Conc. Region }

- Main Roof Immediate Roof & bedding planes is different

How will NPA change with change in h?



→ NPA increases in height & curve.

→ pillar should be able to withstand the stress generated due to stress relaxed zone.

→ beyond 400m, ↑ thickness, hard for B & P. to be maintained

(3.) Pillar Spacing :-

(4) En-masse Caving

↳ high volume caving.

~~BF = HC / ht. of caving ht.~~

$$h_c = \frac{h}{K-1}$$

$h_c \rightarrow$ caving height.

$k \rightarrow$ bulk factor

$h \rightarrow$ ht. of working.

$\boxed{k \rightarrow 1.06 - 1.8}$

→ for Indian geomining condition.

h	K = 1.1 h_c
2.5	25
5	50
7.5	75

as ht. of working increases, height of caving also increases.

$$\frac{2.5}{1.8-1}$$

$$\frac{5}{1.8-1}$$

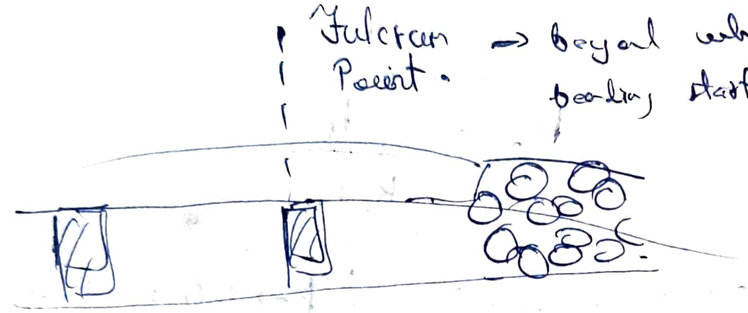
k = 1.8	k = 1.06
3.125	42
6.25	83
9.375	125

$$\boxed{BF = \frac{\text{change in vol}}{\text{original vol}}}$$

④ Nature of caving

① Bulking factor controlled caving.

② Parting factor caving. → larger overhang. → caved material, doesn't touch main roof

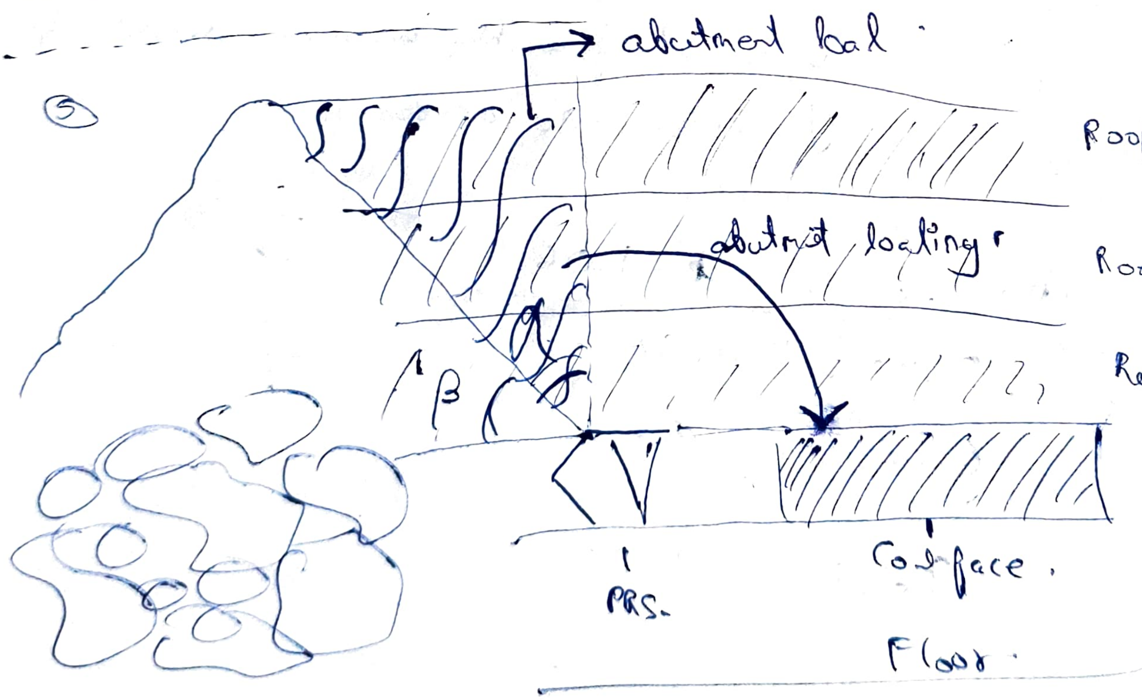


* bulking caving is more favourable..
- more overhang, face spall, support leaching.
- good BF, must propagate above, the caving.

o main & strata weighting

→ abutment load.

⑤



Caved Material.

β → caving angle

α → abutment angle.

• Caving depends on strain, lamination.

• α can't be equal to 0.

* Poor caving, $\beta \downarrow$, then load on pillar $\uparrow$. ($\boxed{\beta > \alpha}$ is a necessity) ***

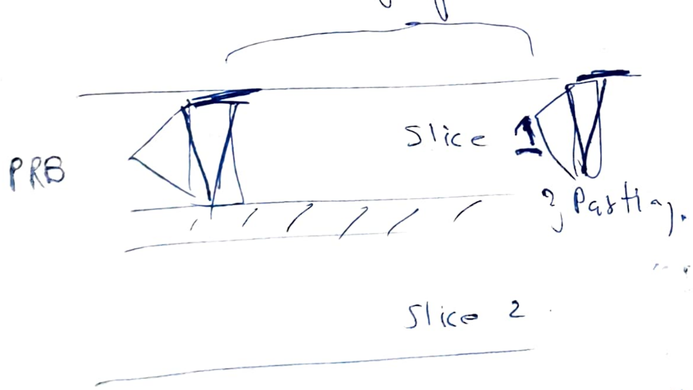
• Great Central Para ⇒ α, β discuss
• Generalized ⇒ ppt.

EXAM POV.

13/11/2025 :-

Problems of mining thick coal seams

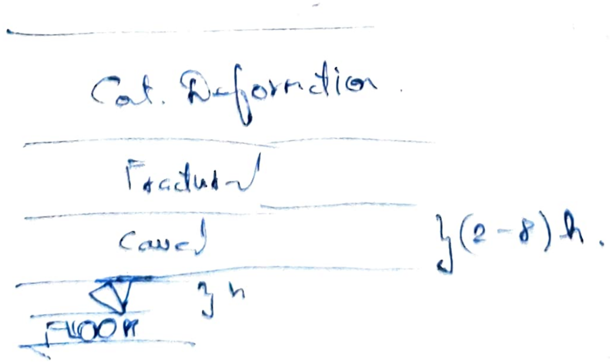
• Multi-seam mining :- for very thick seams.



① Slice I
Slice II ✓

② Slice II
Slice I → back filling

• mine, then stress redistribution
• method selection depend on mining condition, economy.



• Parting should withstand the load due to caving. {* thickness of parting ↓}.

- Thicker seam, high E storage, when released ⇒ coal bumping

- Overriding of Pillar ⇒ at high CI.
  - ⇒ high stress, higher depth

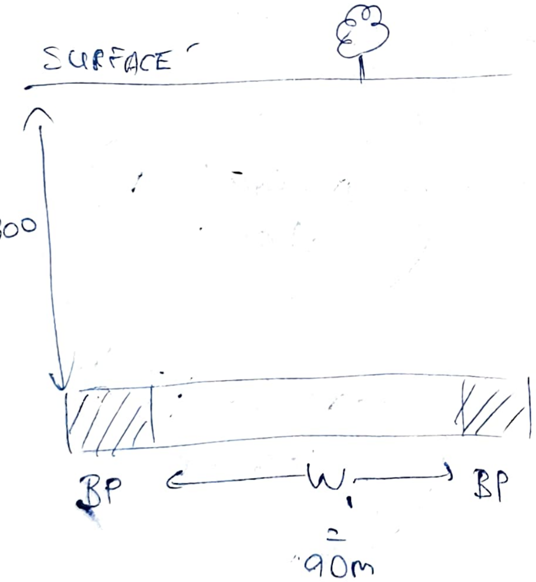
- Sluggish ventilation ⇒ methane layering.
  [thick → more coal in 1 cycle
  → more gas]

- Hooridih.
- 2-5m thk.

- Magnitude of subsidence is high.

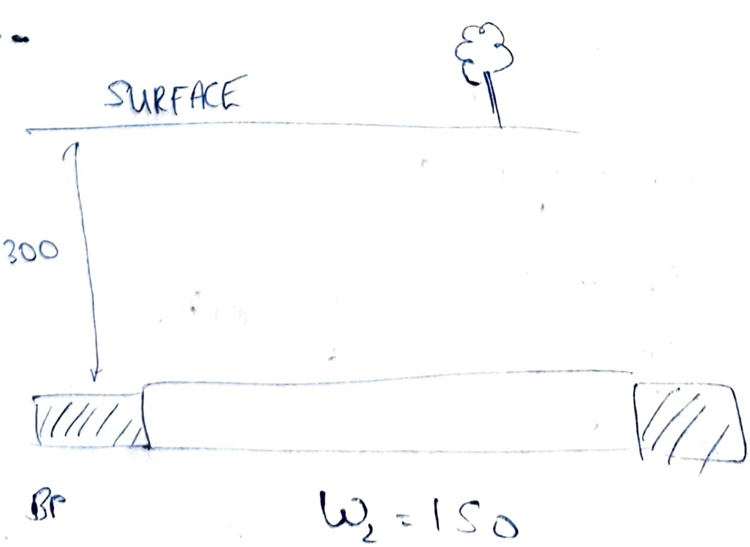
I

w/ Jd



$w/H = 0.3$

II



$w/H = 0.5$

$\Rightarrow$ $\boxed{\begin{array}{llll} W/H & < 1 & \Rightarrow & \text{Sub Critical.} \\ W/H & = 1 & \Rightarrow & \text{Critical} \\ W/H & > 1 & \Rightarrow & \text{Super- Critical} \end{array}}$

NEW $\Rightarrow$ effect of caving doesnt translate to surface $\{W/H = 0.2 - 0.8\}$

$W/H \rightarrow h$ is not considered.

$\rightarrow$ higher h value influences rate of subsidence, higher influence zone.

$\Rightarrow$ W/H define subsidence. extent.
h defines rate of subsidence.

- Geological ~~gradient~~ & constraint and gradient

$\rightarrow$ Extreme characterization & study to be conducted before longwall mining.

15/1/2025 :-

Longwall Mining

- Brief description
- Applicability.
- Design parameter.
- Strata mechanics
- Optimal parameter for designing L/W workings → It is site specific
- Mechanisation / automation in L/W working
- Case Study. {India & Abroad}
- Consideration for designing high capacity L/W.

- PRS width → 1.5 - 2.0m.
- development, extraction.

→ along dip-rise direction / strike direction.

→ stowing not common, mostly. LW with caving.

→ Chain Pillar → with crosscuts → multiple entries → better ventilation (more quantity)
  → flexible
  → Barrier Pillar
  → Yield Pillar → width less. ; extraction ↑, low strength

Maingate → Stage loader, all machineries, so we use chain pillar, face conveyor

→ For free face → support by hydraulic props.

→ Yield Pillar design when we require higher extractions.

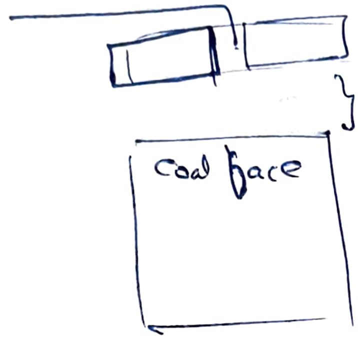
→ Bolter Miner → Development & bolting same time.

→ Salvaging Operation → moving machineries to other panel.
→ 2-3 months } minimize it

 → flencing → bolt, mesh

We prefer wide LW face.; less salvage time.

Bleeder Entry



} Bleeder ~~Entries~~ Pillar

}→ 6-8m.

* Panel Dimension } decides extent of fracture zone.
  Rate of advance }

APPLICABILITY :-

+ thickness uniform in panel.

4-8   2.5

→ Reserve

→ Geology

→ Seam characteristics. , dip (1 in 3), thickness (0.6-6).

8.8 → Chinnar

4.8 → Khottadih.

4.5 → Jhanjra

→ roof and floor strata.

→ geological anomalies.

Shearer } requirement??
Plough }

Roof & floor Strat.

→ Roof easily cavable.

→ Floor → floor heave & penetration

→ + load transferred to it through face and roof.

Strain } defines massive roof.
Thickness }

⇓

① Induced blasting.

② Size of opening such that fracture & its propagation in behind face.

16/1/2025:-

- Negation of geological anomaly is time consuming

- Cover Depth → >200m

B&P → deep, pillar size ↑, recovery ↓
→ ventilation has to reach multiple faces
→ huge quantity required.

Width → 360 - 500m

Length → 3500 - 7000m

India → 250m
2.5 km.

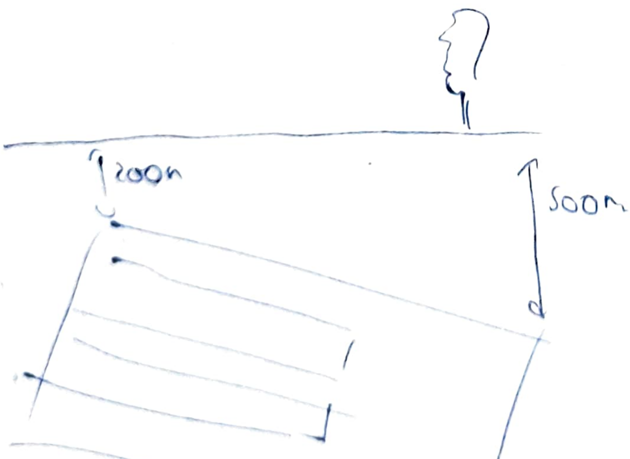
- Suppose 2 km length and at 600m, fault → extract till 550m, 50m coal left for support.

- Strength of Coal:-
req. cutting → soft to medium.

→ strata control pov
→ production pov.

- Gestation Period → 1-5 years for single panel

NEW → $\frac{\omega}{H}$ → 0.2 - 0.8



$\frac{150}{200} = 0.75$ → minimum

$\frac{\cancel{150}^{3}}{\cancel{500}_{10}} = 0.3$

- $\frac{W}{H}$ → we consider minimum depth as max depth underestimates

• NEW, no consideration of the strata above.

Subsidence

- Deformation
- Strain

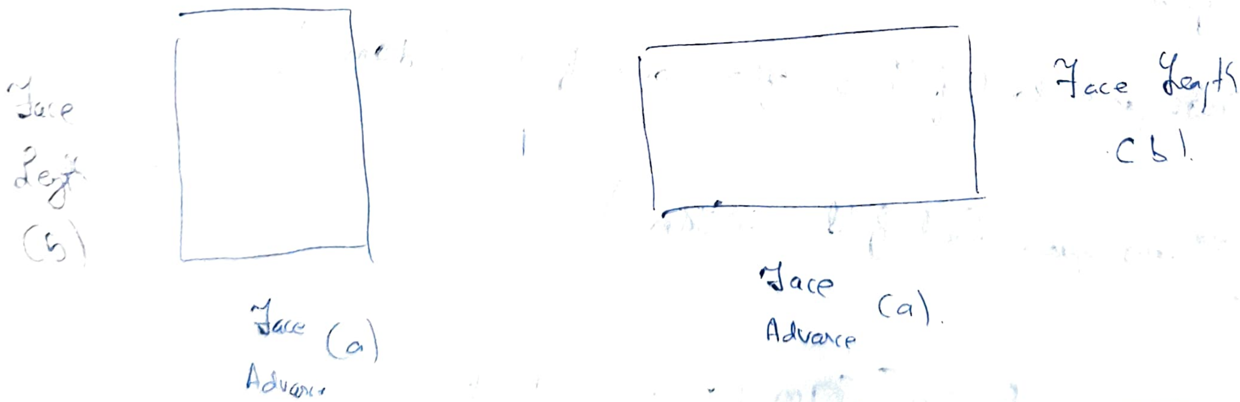
Advancing → support of gateroad throughout.
→ blind heading development.
→ efficiency ↓
→ sluggish ventilation.

→ geology cannot be explored.
⇓
may halt face advance

Retreating :-

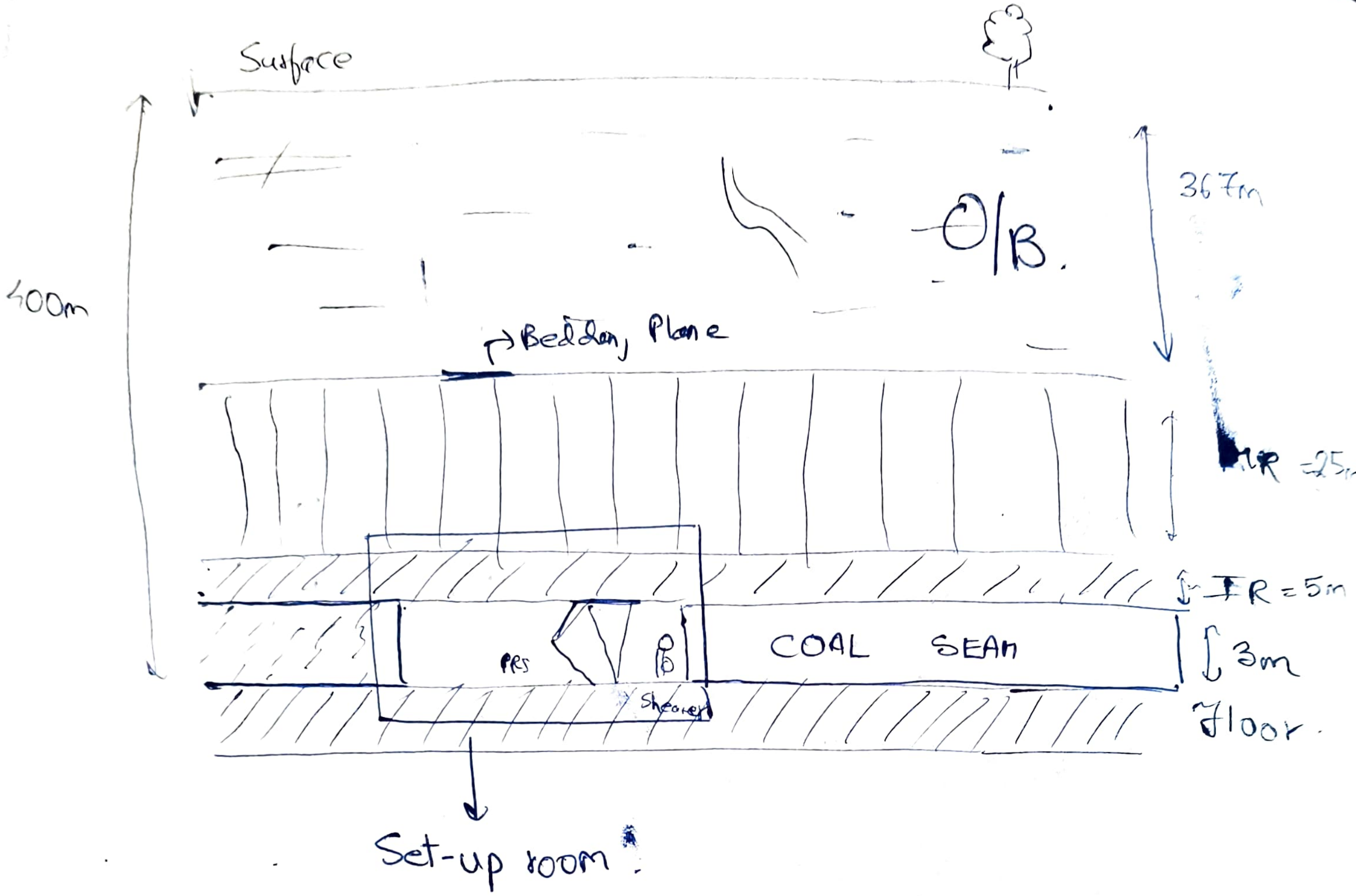
* Longwall Requirement → face has to keep moving.

Different Parameters of LW



① Local fall.
② Subsequent local fall.
③ Main fall.
④ Periodic caving.

• Convergence Rate ↑
⇓
~~elastoplastic~~ → elastoplastic.
elastic



- **Local fall** → rock tries to fall from IR.
  → plastic to elasto plastic deformation. (20m)

Now, plate converts to cantilever.

- **Subsequent local fall** → after local fall (30m).

- **2nd Subsequent local fall** → (38m.)

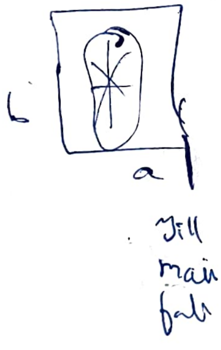
- **Main Fall** ⇒ (60m) → Main Roof.
  - ↳ load transfers to face
  - ↳ main weighting.

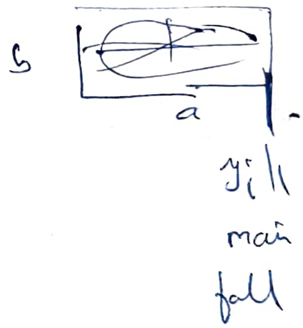
• Periodic Caving → load transfer is periodic weighting.
→ subsequent fallings of the main roof

* The spans depend on
① Rate of advance.
② nature of roof.

★ Kirchoff's Plate Theory

• fracture propagate along longer dimension {a or b}.
• then it forms star.



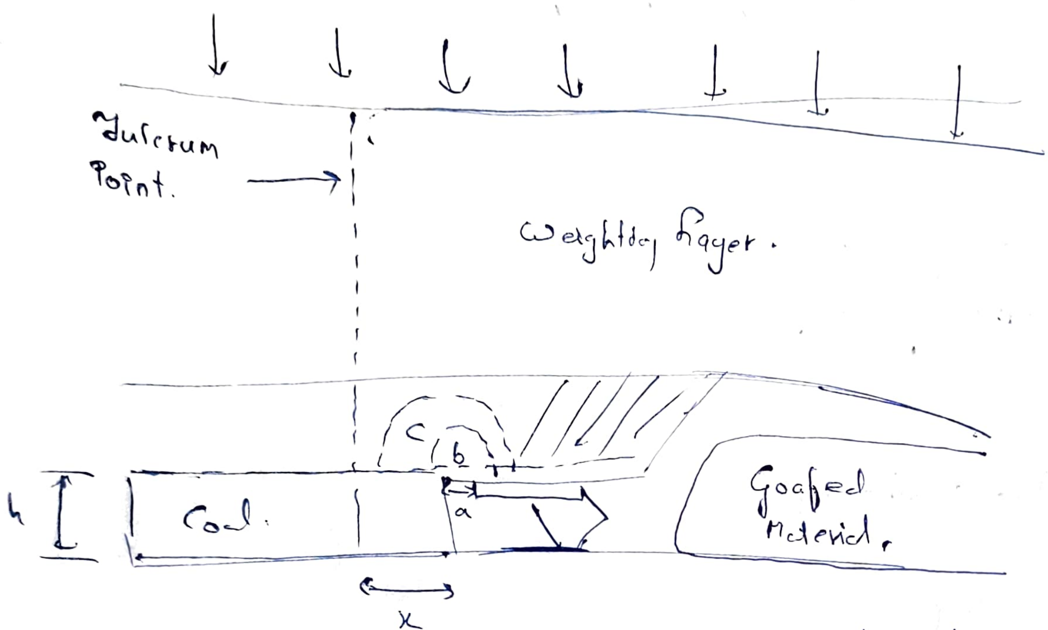


$a/b < 1$ ⇒ more frequent caving
$a/b > 1$ ⇒ longer time to cave

20/11/2025 :-

For $a/b < 1$ ⇒ caving is faster
For $a/b > 1$ → caving is delayed
} all other parameters are fixed, we consider only dimension.

1st :-



b → guttering of roof

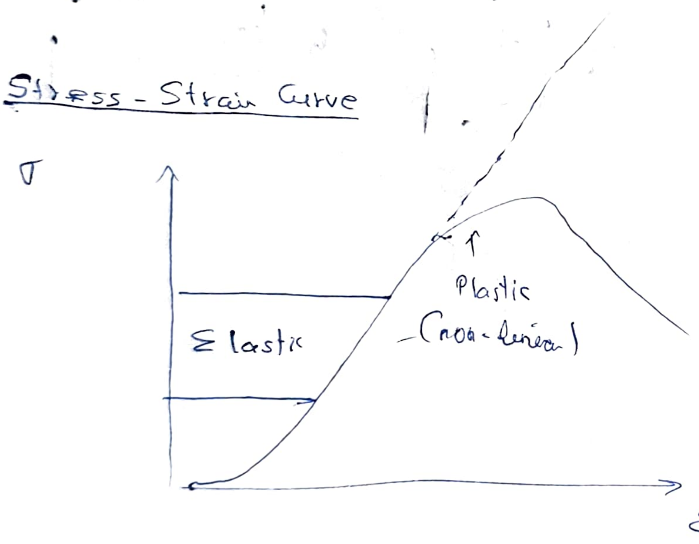
c → spalling drives cavity,

a → tip to face distance

⇓

roof cavity ⇒ worst case scenario

Stress - Strain Curve



- Face length ↑, ~~BL close~~ fall faster
- Face length ↓, BL further

2nd :-

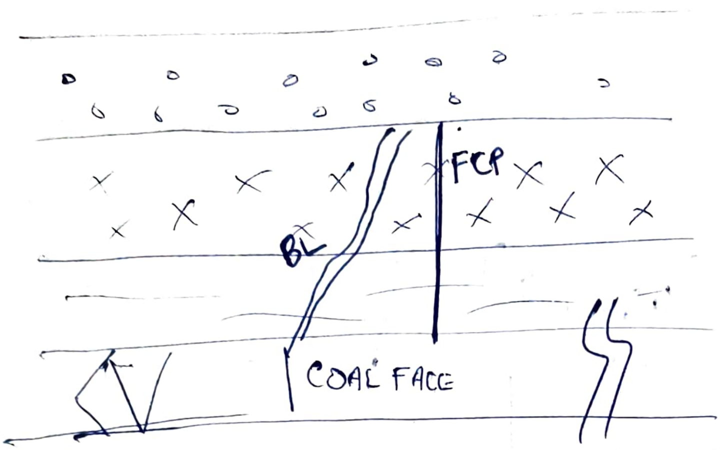
FP → indication of loading of roof.

Elastic ⇒ Bending.

Elastoplastic ⇒ loading to face

→ strain energy released

Case1 :- Face length >> Span of main Fall.



BL → Breaker Line

FCP → Fulcrum Point.

Reach BL ⇒ mainfall will occur.

→ more pillar spalling.

* ~~Ideal Case when BL is behind~~

* BL behind, larger span, may face wind blast

Case 2 :- Face length < Span/main fall.

India Scenario, Cavin Zone = 12-15 H

→ Difficult to cave
→ high bulk factor.
→ 80-85% Sandstone

Low BF ⇒ more volume to fill
→ higher cave zone.

* Compacted goaf can be considered as regional support.

Abutment Stress Distribution :-

→ Flat Jack → 6-compon (σ_{xx} σ_{yy} σ_{zz} τ_{xy} τ_{yz} τ_{xz})

→ Hydrofracturing → all at once

→ Insitu stress, induced stress.

$$\boxed{\sigma_H = \frac{\nu}{1-\nu}\sigma_v + \frac{\beta E G}{1-\nu}(1000+H)}$$ or $$\boxed{\sigma_H = \frac{\nu}{1-\nu}\sigma_v}$$

β = thermal expansion coefficient.

G = Geothermal gradient.

ν → Poisson Ratio

H → depth of working

σ_H → mean horizontal stress.

→ $\dfrac{\sigma_{h_1} + \sigma_{h_2}}{2}$

→ avg of major and minor horizontal stress.

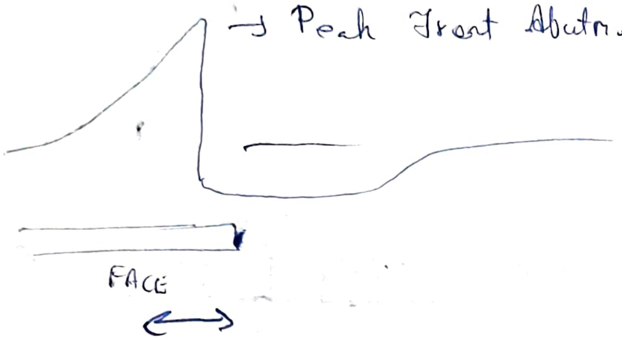
Abutment → solid takes stress

→ goaf takes stress when it is compacted

T-Junction → more stress

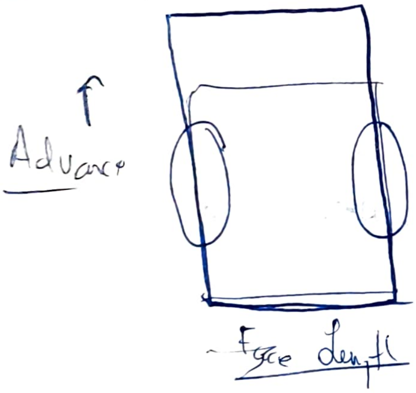
→ more superposition of stress.

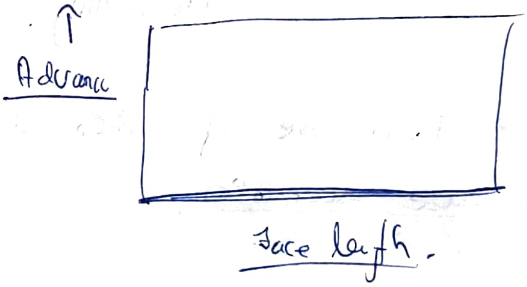
$$\boxed{\text{Zone of influence} = \sqrt{20} \cdot a}$$



$\boxed{2-5m}$ → from face, load is observed.

22/1/2025 :-

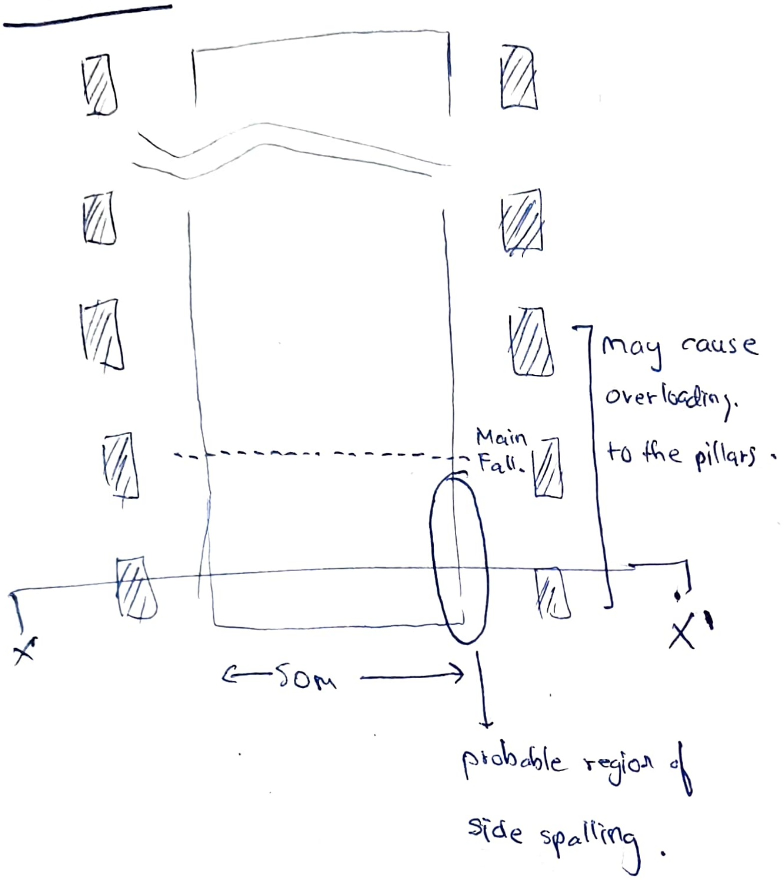


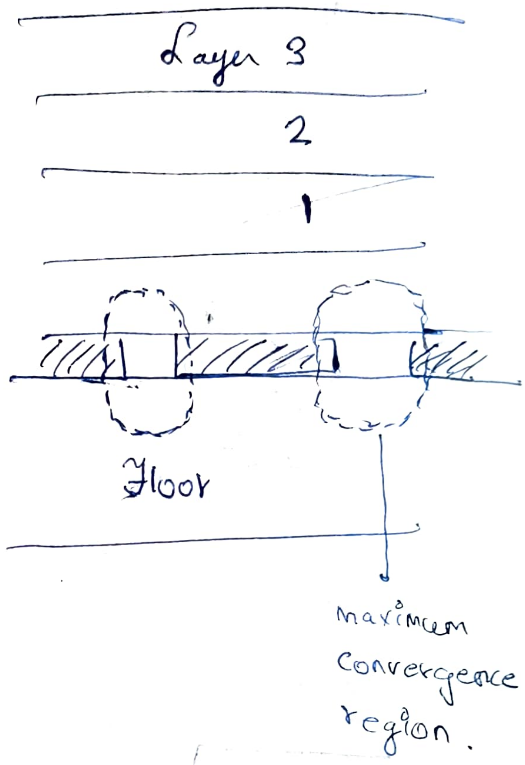


→ Load at gateroads

→ more convergence

Cond^n 1 :-

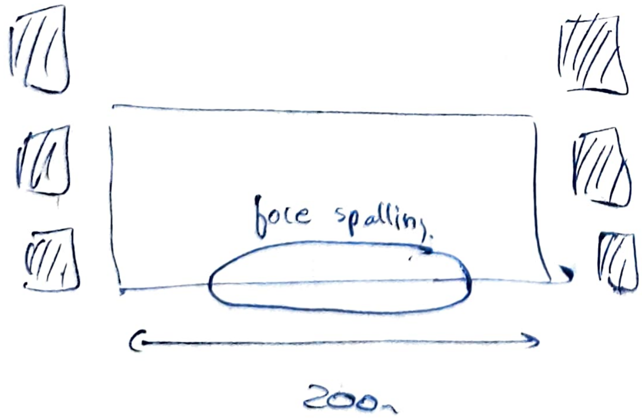




⇒ this cond^n at massive roof, when it is hard to cave. This is the **worst case scenario**

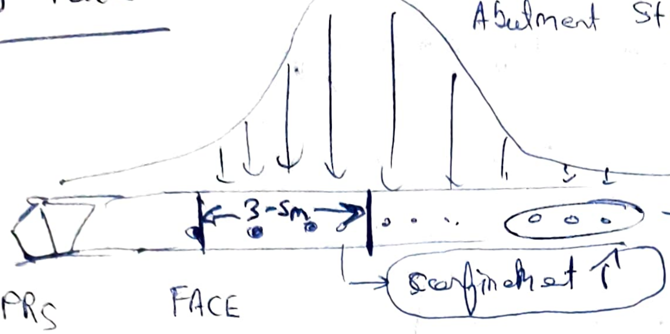
Cond^n 2 :- Load at face → may cause face spalling.

• max^m load to face and PRS



• Across Face

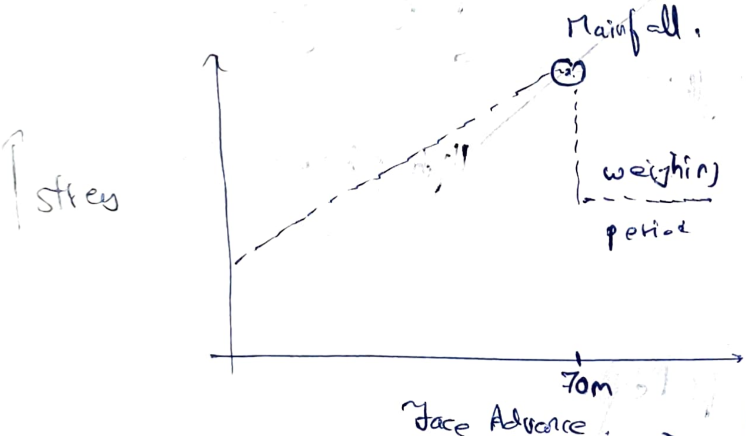
Abutment Stress Distribution.



?!

it decrease, in-situ, why

• Due to confinement, we don't get uniform stress distribution



→ monitor peak load with face advance. (confining & vert. zone.)

→ after maxm. stress, stress ↓

→ due to plasticity.

this cycle will be repeated till we finish

→ load transfer to face → face spalling.

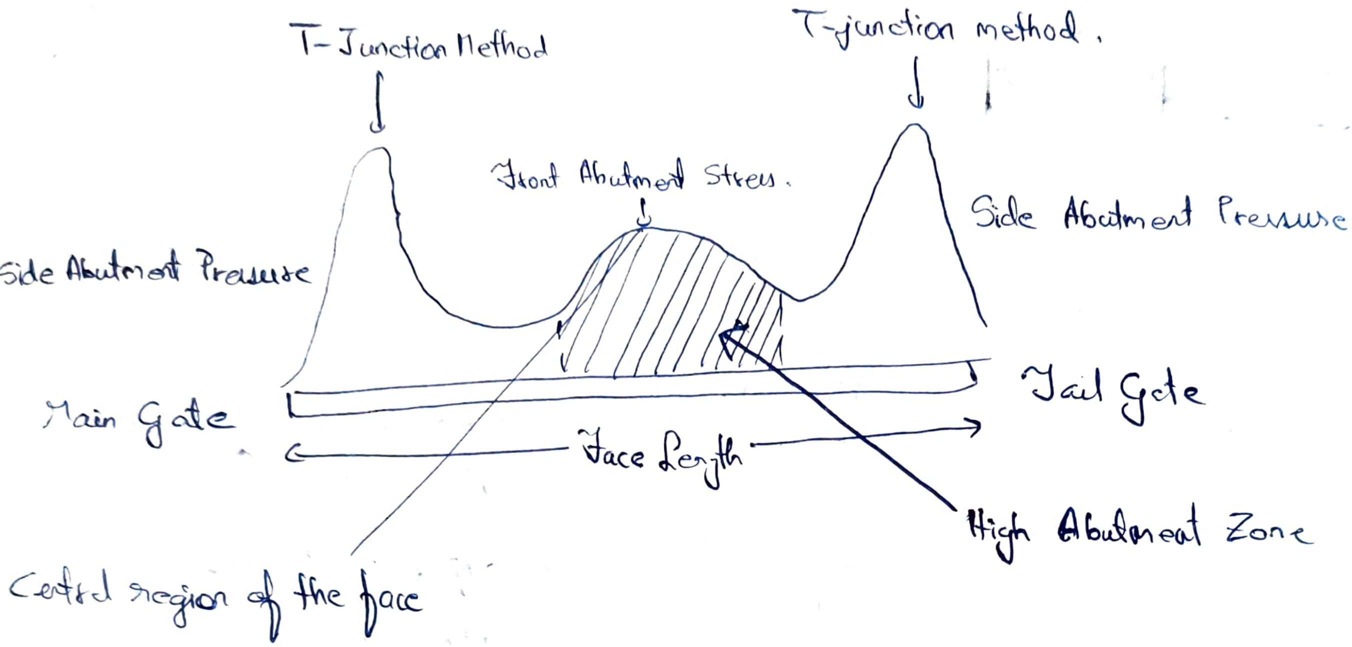
→ main fall → load is now transfered

→ face length ↑, its domain of effect also ↑, so we have to find optimum.

→ control spalling of face after main fall, if not, it will go deeper

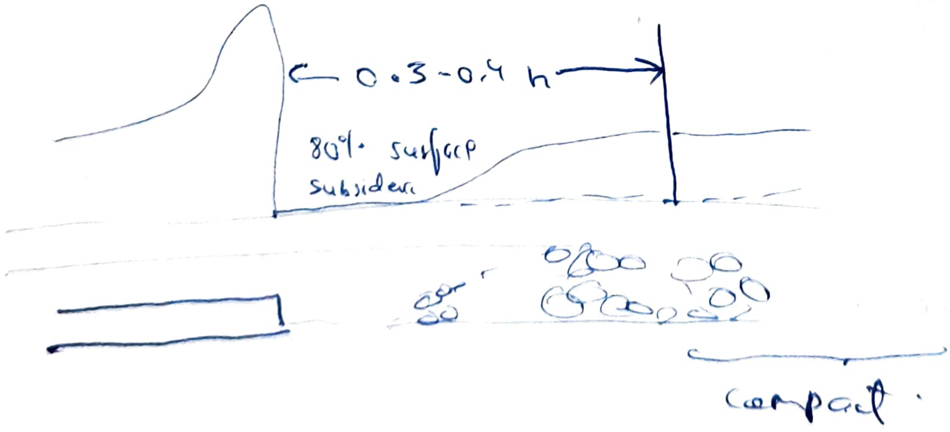
2.) Along the face.

- Stress redistribution ~~ahead long wall~~ along length of face

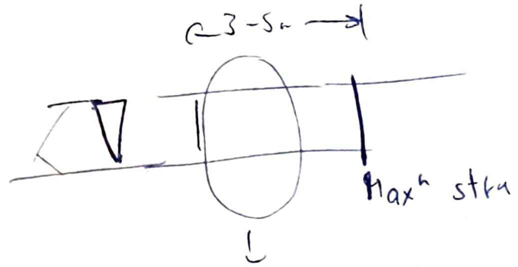


Centre → due to bending of plate
T-junction → due to superimposition

Cover Pressure Distance, ⇒ goaf compact, it takes load
→ return to in-situ stress



- zero stress near face, as goaf doesn't touch roof

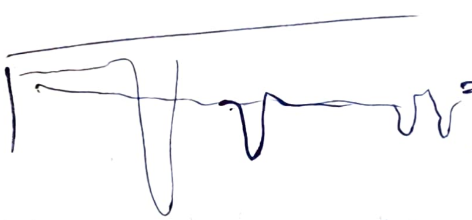


low confinement, we can't have high stress, zone of max. damage,

22/1/2025:-

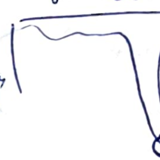
Weighting at the Face :-

- strata condn, geomining parameters.
- Massive → high thickness high strength.
- Thick Roof → low strength



⇒ caving in pockets
→ graded failures.

▼ En-masse caving.
⇒ high volume caving.
→ sudden peak convergence



Yield Point → Bleeding of support

Massive Roof → best into do induced blasting

↱ Support
↳ face convergence } define weighting characteristic of LW.

Peak Stress ⇒ Cave ⇔ Weighting
&
Reflected as damage ⇒ spalling

27/1/2025 :-

- Before each major fall, 3 conditions,

→ Normal Period → less convergence
→ Peak Stress
→ Weighting Period

* Max^m convergence at centre

→ Worst Case Scenario

- Cavity ahead of powered support → collapse
  ⇒ secondary handling
  → first, we arrest cavity.

- grouting not possible, cavity too thick ⇒ wooden props.

- Convergence ↑ ⇒ Face spalling
  ⇒ Tip to face distance ↑.

Roof → laminated
Roof → massive

* Powered support weak, then cavity / fulcrum advances away

- Bend from fulcrum → compression
  → try to deform in other directi
  → but there is compress from surround
  ⇒ roof converge

- Guttering & Roof Cavity } both in front of support.
  Guttering ↓ local (initial face)
  Roof Cavity ↓ main.

⇒ Support system should match with strata condⁿ.

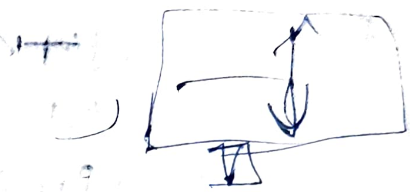
→ Tensile Fracture ⇒ breaker line.

→ Centre ⇒ due to cantilever

Laminated → proper support

Massive → breaker line just behind support

→ too much behind ⇒ wind blast.



→ Shortwall → caving delay. → induced blast.

→ Longwall.

* Thickness of main roof excessive → fracture difficult to propagate.

→ induced blasting

→ or increase face length.

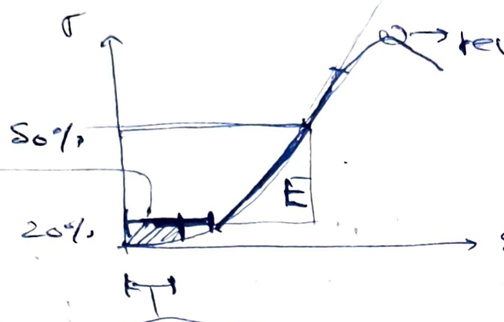
→ Caving Shortwall → delayed caving.

→ Bridging Shortwall → no caving → ↑ face width.

→ (condⁿ. of caving not satisfied).

29/1/2025 :-

$\boxed{\sigma_1 - \sigma_3 \rightarrow \text{differential / differentiated stress}}$



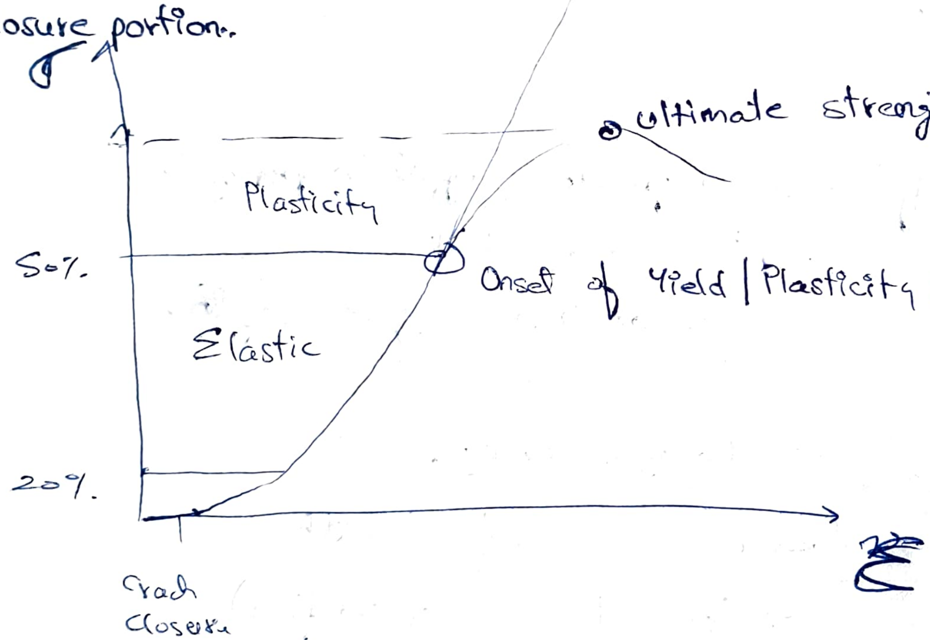
→ reverse ⇒ indicate failure
⇒ highly damaged.

20% σ_c

Pores Close ⇒ Elastic ⇒ Inelastic → Peak → failure

→ to close pores.

→ crack closure portion.



ultimate strength

Onset of Yield | Plasticity (~ 70-80% of σ_c)

∴ After yield, crack damage propagates, then failure.

- EIZ → Excavation Influence Zone
- EDZ → Excavation Damage Zone → from withdrawal of liner → plasticity zone
- HDZ → Highly Damaged Zone

→ differentiated based on stress distribution.

→ analysis during peak stress period.

Face | Plastic → Elastic → Intact

Damage Zone

- Normal Cond^n
- Peak Cond^n
- Weighting. Cond^n.

→ face guard → provides confine

→ prevent face spalling

Face

→ Spalling from centre to top portion, propagation,

→ because floor is strong rock, which top is coal

more confinement | less confinement

| ◦ Tensor | + | Face Damage & Spalling |

- Face advance should not stop, if so, loss + spalling propagate + ground control difficulty.

Roof Separation Index (RSI)

→ to determine thickness of intermediate, main roof.

→ identify layers of rocks.

| Caving Zone → (12-15) × Working Ht. |

→ 3x15 → 45 → till this we use RSI for interoolide | main roof

Parameters :- UCS, RQD, ACL

ACL ↓ avg. core length.

Gtotal →

Parameter	Unit/Rating	Rating Distribution				
① UCS.	MPa	<5	5-25	25-50	>50	
	Rating.	1	2	4	7	
② RQD	%	<25	25-50	50-75	75-90	>90
	Rating.	3	8	13	17	20
③ ACL	cm	<5	5-30	30-100	100-300	
	Rating.	5	10	20	25	
④ GW condⁿ / wetness of strata.	Condⁿ.	Dry	Water formation			
	Rating.	10	0			
⑤ Bed thickness.	m	<1.5	1.5-2.5	2.5-3.5	3.5-4.5	>4.5
	Rating.	-28	-16	-12	-5	0

⑥ Rock formation of bed.	Rock Type	Sandstone	Others (Shale, Coal, clay, shaly coal)
	Rating.	0	-25

Rock Type	Thickness	RQD %	σ_c MPa	σ_t MPa	Density kg/m³	Emod. GPa	ACL cm.
Coal	3	40	23.16	1.54	1400	2	7.70
Cgsst, GW	1.3	44.	12.46	1.07	2050	3.86	8.67
Cgsst, GW, Pebble	4	93	8.68	0.9	2040	2.69	29.27
Cgsst, GW Pebble	3.7	77	8.68	0.9	2040	2.69	13.70
Cg to Fg sst	6.53	94.	12.93	1.15.	2060	4.01	19.2
Cgsst, GW Pebble	12.49	78	8.68	0.9	2040	2.69	16.6
Grey & ~~continuous~~ carbonaceous clay	2	50	12.46	1.07	2050	3.86	7.7
Cgsst, GW	8.28	80.3	12.46	1.07	2050	3.86	23.65.
Grey & carbonaceous clay.	7.66	50	12.46	1.07	2050	3.86	7.7
Cgsst GW	8.54	69	12.46	1.07	2050	3.86	14.1
Cgsst GW	7.3	85	12.46	1.07	2050	3.86	32.14
OB	150.5						

→ Check for bedding plane.

RSI

Coal

Bedding -6 Plane — IR = 30.02 m

Bedding .10 — MR = 15.94 m.

∴ Qualify each layer ⇒ to be taken as IR or MR.

→ take weighted avg ⇒ $\dfrac{\sum t \times (RQD \text{ or } \sigma_c \text{ or } \sigma_t)}{\sum t}$.

RSI = Sum of (all 6)

→ Borehole data ⇒ Intact Rock

⇒ to be converted to rockmass

$$\boxed{\sigma_{c_{\text{Rock mass}}} = 0.5 \times \sigma_{c_{\text{Intact Rock}}} \times RQD}$$

→ Same for tensile

0.5 → Scale Factor.

$$\boxed{E = (0.31 \times \sigma_{c_{IR}})\ GPa.}$$

- **Plane Strain and Plane Stress**

31/1/2025 :-

$IR \rightarrow 30.02$

$MR \rightarrow 15.94$

	UCS	RQD	ACL	GW.	Bed Thickness	bed formation	RSI
Coal	2	8	10	10	-12	-25.	-7
Cgsst, gw	2	8	10	10	-28	0	2
Cgst, gw Pebble	2	20	10	10	-5	0	37
Cgsst gw Pebble	2	17	10	10	-5	0.	39
Cg to Fg sst	2	20	10	10	0	0	42
Cgsst, gw Pebble	2	17	10	10	0	0	39
Grey. carbo. clay	2	13	10	10	-16	-25	-6
Cgsst, gw	2	17	10	10	0	0	39
Grey, carbo. clay	2	13	10	10	0	-25	10
Cgsst, gw	2	13	10	10	0	0	35
Cg sst gw	2	17	20	10	0	0	39
OB							

IR MR

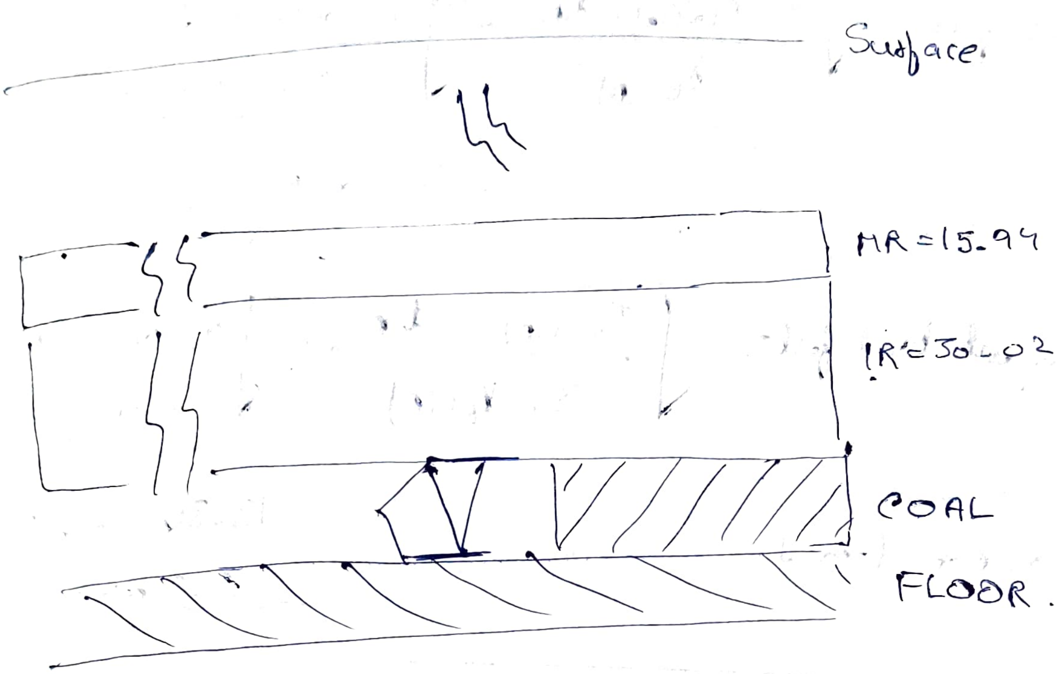
IR = 30.02m

MR = 15.94m

	Thickness
IR	30.02
MR	15.94

Layer	Thickness	ρ	σ_c (intact Comp. Str.)	σ_t (intact)	σ_c (RM)	σ_t of RM
Coal.	3	1400				
IR	30.02	2045				
MR	15.94	2050				
OB	166.34	2050.				

Layers	Thickness	ρ kg/m³	RQD	σ_c (IR)	σ_t (IR)	σ_c (RM)	σ_t (RM)
Coal	3	1400	90	23.16	1.54	4.632	0.308
IR	30.02	2045	80	10.02	0.97	4.008	0.388
MR	15.94	2050	66	12.46	1.07	~~4.095~~ 4.118.	0.352
OB	166.34	2050	76	12.46	1.07.	~~4.409~~ 4.739	0.4066



i.) Local fall
ii.) Subsequent Local fall
iii.) Main fall
iv.) Periodic caving

} Beam Model
} Cantilever Model.
} Beam Model
} Cantilever Model

$\boxed{\gamma_e \to MPa/m}$

Models Used:
→ ① Beam ~~de~~ model $\Rightarrow \sqrt{\dfrac{2 \times \sigma_{tm} \times t}{\gamma_e}}$
→ ② Cantilever model. $\Rightarrow \sqrt{\dfrac{\sigma_{tm} \times t}{3\gamma_e}}$

$\boxed{\text{LW} \to \text{Plane Strain.}}$

$\gamma_e \to MPa/m \to$ Eff. unit weight $\Rightarrow (\rho \times g \times 10^{-6})$ MPa/m.

$\therefore$ $\gamma_e(IR) = 0.02046$ MPa/m ($g = 10$ considered)
$\gamma_e(MR) = 0.0205$ MPa/m

Local fall

Span of local fall $= \sqrt{\dfrac{2 \times \sigma_{tm(IR)} \times t_{IR}}{\gamma_{E(IR)}}} = \sqrt{\dfrac{2 \times 0.3625 \times 30.01}{0.02046}}$

$= 32.62 m$

Span of subsequent local fall $= \sqrt{\dfrac{\sigma_{tm}(IR) \times t_{IR}}{3 \times \gamma_E(IR)}} = \sqrt{\dfrac{0.3625 \times 30.02}{3 \times 0.02046}}$

can be more than 1

$= 13.318 + 32.62$

$= 45.94$

Span of main fall $= \sqrt{\dfrac{2 \times 0.352 \times 15.94}{0.0205}} = 23.4 m$

$= 23.4 + 45.94$

$= 69.34$

Periodic Caving Span $= \sqrt{\dfrac{0.352 \times 15.94}{3 \times 0.0205}} = 9.55$

$= 69.34 + 9.55$

$= 78.89.$

- Not considered → Rate of face advance & face width consideration } → Mining Operation.
  / no consideration of 3D, as it is 2D based

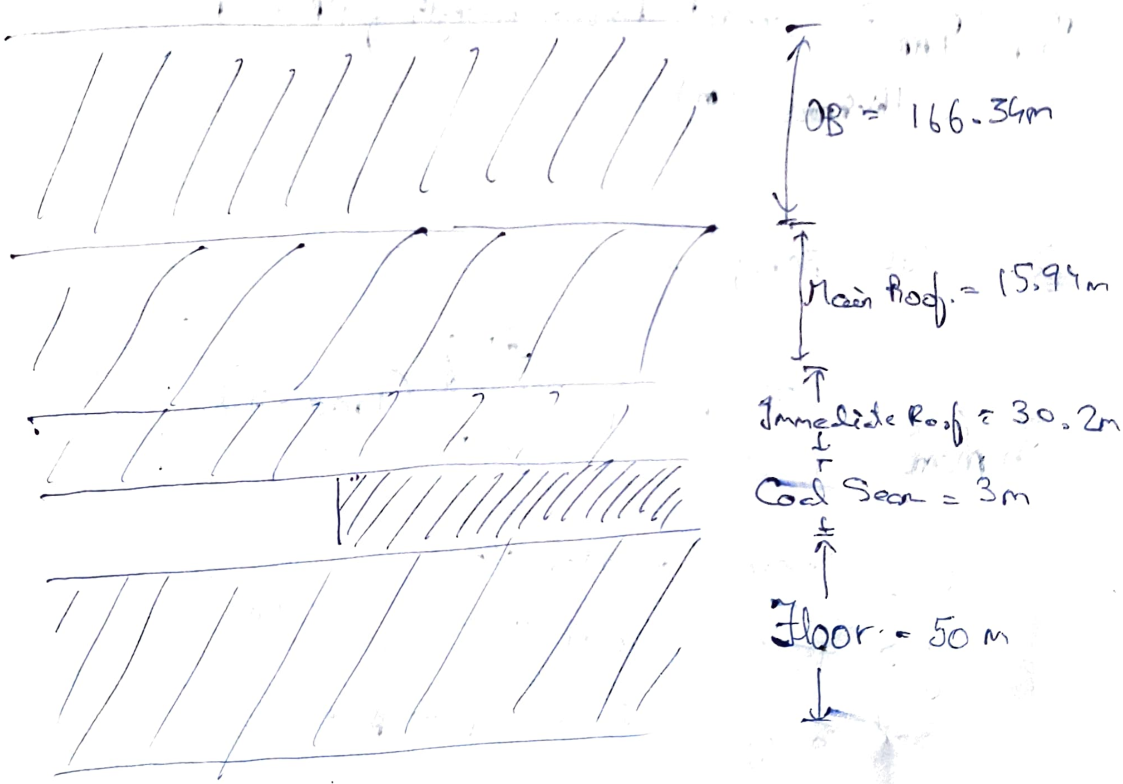
69.34 - 32.62 = 36.98

$\frac{36.98}{13.318} \approx 3$. subsequent local falls

12|2|2026:-

Conceptual / Sequence of Caving:-

* Relation taking face length effect



$$W_S = 324 \times (F.L)^{-0.43} \times D^{-1.38} \times \sqrt{\frac{\sigma_{MIR} \times T_{IR}}{\gamma_{IR}}} \times \sqrt{\frac{\sigma_{MMR} \times T_{MR}}{\gamma_{MR}}}$$

$W_s \rightarrow$ weighing span / main fall span (m)

$\gamma \rightarrow$ unit cut

$FL \rightarrow$ face length (m)

$D \rightarrow$ depth of working (m)

$\sigma_{MIR} / \sigma_{MMR} \rightarrow$ Equivalent tensile strength of MR or IR

$T \rightarrow$ thickness (m).

Q. $FL = 150m$;

$$\sigma_{h\,mean} = \frac{\nu}{1-\nu}\sigma_v + \frac{\beta E G}{1-\nu}(1000 + D).$$

$\sigma_{h\,mean} \rightarrow$ mean horizontal stress

$G \rightarrow$ geothermal gradient $= 0.03\ °C/m$

$\beta \rightarrow$ co-efficient of thermal expansion.

$\rightarrow 8\times10^{-6}\ /°C \rightarrow$ ~~COAL~~ SANDSTONE

$\rightarrow 30\times10^{-6}\ /°C \rightarrow$ ~~SANDSTONE~~ COAL

$E \rightarrow$ Young's Modulus $= 3.1\,GPa = 3.1\times10^{3}$

$\nu \rightarrow$ Poisson's Ratio $= 0.25$.

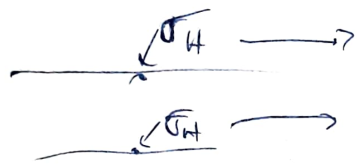
$\sigma_v \rightarrow$ Vertical Stress $= \gamma D$; $\gamma = 0.023\ MPa/m$

$$\sigma_{MIR} = \frac{(\sigma_{TIR} + \sigma_{hmean\,IR})}{100} \times RQD.$$

$\sigma_{tIR} \rightarrow$ equivalent tensile str. of IR. $\Rightarrow$ ~~(INTACT ROCK)~~. (ROCK MASS)

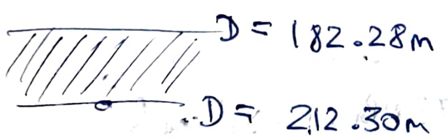
$\sigma_{hm\,IR} \rightarrow$ mean horizontal stress corresponding to IR

RQD $\rightarrow$ Rock Quality Designation.



Take average

ii) IR



$$\sigma_{hm}(top) = \frac{0.25}{0.75} \times (0.023 \times 182.28) + \frac{30\times10^{-6} \times 3.1\times10^{3} \times 0.03}{0.75} \times (1000 + 182.28)$$

$$= \cancel{5.795\,MPa} = 2.57\,MPa$$

$$\sigma_{hm}(bottom) = \cancel{6.137\,MPa}\ 2.83\,MPa$$

$$\sigma_{hm\,(avg)} = \frac{\sigma_{hm_t} + \sigma_{hm_b}}{2} = 2.7\,MPa$$

$$\sigma_{MIR} = \frac{(2.7 + \cancel{0.97}\ 0.388)}{100} \times 80 = \cancel{2.93\,MPa}\ \boxed{2.47\,MPa}.$$

ii) MB

Top $\Rightarrow D = 166.34$

Bottom $\Rightarrow D = 182.284$.

$\boxed{E \rightarrow \sigma_E = 0.31 \times \sigma_c}$

$$\sigma_{hm}\,(top) = \frac{0.25}{0.75} \times (0.023 \times 166.34) + \frac{8\times10^{-6} \times \cancel{3.7}^{3.86} \times 10^{3} \times 0.03 \times (1000 + 166.34)}{0.75}$$

$$= 2.72\,MPa$$

$$\sigma_{hm}\,(bottom) = 2.86\,MPa.$$

$$\therefore \sigma_{hm} = \frac{2.72 + 2.86}{2} = 2.79\,MPa$$

$$\boxed{\sigma_{MHR} = \frac{(2.79 + 0.352)}{100} \times 66 = 2.073\,MPa.} \rightarrow \text{tensile}$$

$$W_s = 324 \times (150)^{-0.43} \times (212.48)^{-1.38} \times \sqrt{\frac{2.47 \times 30.02}{0.02046}} \times \sqrt{\frac{2.073 \times 15.94}{0.0205}}$$

$\boxed{W_s = 35.77\,m}$

$\boxed{(\gamma \times g \times 10^{-6})\,MPa/m}$

* Variation of D → all of σ_{hm} changes
FL → only W_s value varies.

FL↑ → High abutment Zone Increases depth wise

Face Height ⇑ ⇒ stiffness ↓

Roof Thickness ↑ → face stress conc. ↑ → face spalling

$\boxed{\text{Roof thickness Ratio} = \frac{MR}{IR}}$

★ Cavity Prone → Soft IR

17/2/26

$$S_v = 0.0014\, D^{0.54}\, h^{1.96}\, F_L^{1.36}\, \sigma_{CIR}^{0.76}\, \sigma_{CMR}^{6.04}\, \sigma_{CCS}^{-0.34}\, T_R^{0.17} \left[\frac{E_{IR}\, E_{IF}}{E_{C}^{2}}\right]^{-0.47}$$

S_v → spalled vol. of coal, m^3

D → depth

h → mining height

F_L → face length.

σ_{CIR} → σ_c of IR

σ_{CMR} → MR

σ_{CCS} → coal.

T_s → thickness ratio of caving layers, $\frac{T_{MR}}{T_{IR}}$

T_{MR} → MR.

T_{IR}

E_{IR} → YM of IR

E_{IF} → YM of floor.

E_{C} → YM of coal.

* Face Damage Severity Classification :-

→ weighting period ↑ → more face spall ⇒ weighting character

→ coal strength, ht. of extraction → 2 variables → coal character.

σ_c, σ_t

Structural variable.

* Weighting, Spalling and Face Advance Rate

→ identify crit parameters,
→ evaluate spall potential.
→ estimate main weighting span,
→ study face progress ⇒ quantify face advance, normal + weighting period.
→ develop relation.

Drop Rate Face Progress: $\boxed{DRFP = 25.583 + 0.092\,S_c + 0.146\,W_s}$

Web Cut of Shearer → 0.8 × Wt → 0.8
→ 0.6 - 0.7.

Category of face dmg.		DRFP (%)	Class of severity of the face damage	Impact of face productivity
I	Low	< 45	Low	Nil.
II	Mod	≥ 45% & < 70%	Moderate	Remains Manageable
III	High	> 70%.	High	Significant

Severity class